FMF-AI

FORMAL METHODS AND FOUNDATIONS
OF ARTIFICIAL INTELLIGENCE

EGER 5–7 June 2025

PROGRAM

Head of the Program Committee
Gergely Kovásznai

Members of the organising committee
Gábor Kusper, Ádám Kovács, Ágnes Nagyné Bertalan, Ede Troll, Erzsébet Tóth, Péter Takács, Péter Szigetváry, Roland Király, Tamás Balla

Organizer:



Thursday, June 5

08:00 - 13:00 Registration in Main Hall of Building C
Eger, Leányka str. 4., Building C

09:30 - 10:00 Opening Ceremony, C-124

10:00 - 10:50 **Plenary talk** (Chair: Olivér Hornyák), C-124
Tibor Tajti – *From Compute to Autonomous Superintelligence: A Quantitative and Formal Perspective*

10:50 - 11:20 Coffee-break, C-119

11:20 - 12:10 **Plenary talk** (Chair: Olivér Hornyák), C-124
Peter Drotár – *Voice as Data: AI-Based Acoustic Analytics for Clinical Classification*

12:10 - 13:00 **Plenary talk** (Chair: Olivér Hornyák), C-124
Isabela Drămnesc – *Teaching robotics, the AiRobo project*

13:00 - 14:00 Lunch in Leányka Bisztró in Building C*
*Eger, Leányka str. 4., Building C**

14:00 - 15:40 Sections

15:40 - 16:20 Coffee-break, C-109

16:20 - 18:00 Sections

18:00 - 22:00 Dinner in Leányka Bisztró in Building C*
*Eger, Leányka str. 4., Building C**

Section A
Machine Learning and Data Science
Chair: Csaba Szabó, C-124

14:00 - 14:20 Salam Allawi Hussein and Sándor R. Répás – *A Federated Learning Approach to Network Event Classification*

14:20 - 14:40 Nasraldeen Khleel and Károly Nehéz – *Code Smells Detection Based on an Automatic Machine Learning Tool*

14:40 - 15:00 Isabela Dramnesc, Erika Ábrahám, László Antal, Péter Csóka, Nikolaos Fachantidis, Tudor Jebelean, Konstantinos Papadopoulos and Gábor Kusper – *Artificial Intelligence based Robotic Applications for Higher Education*

15:00 - 15:20 Benedek Szabolcs Csüllög and Máté Tejfel – *Using GNN for Refactoring P4 Programs*

15:20 - 15:40 Marek Ružička, Martin Štancel, Miroslav Imrich and Dmytro Havrysh – *Physics-Informed Neural Networks for Acoustic Wave Propagation*

15:40 - 16:20 Coffee-break, C-119

Section A
Machine Learning and Data Science
Chair: Peter Drotár, C-124

16:20 - 16:40 Olivér Hornyák – *An LSTM Approach for Fault Prediction*

16:40 - 17:00 Zijian Győző Yang, Lili Anna Stajer and Gergely Lukács – *Under the Hood: An Inside Look at PULI Models*

17:00 - 17:20 Ming Hu and Gergely Kovásznai – *Web-Based Facial Expression Recognition Using Hybrid Deep Learning*

17:20 - 17:40 Emira Mustafa Moamer Alzeyani and Csaba Szabó – *Predictive Modelling for Project Completion Time Using Long Short-Term Memory Neural Networks*

17:40 - 18:00 László Kovács, Enikő Palencsár and Péter Bán – *Efficiency Testing of OpenSet Learning Methods in Image Classification*

Section B
Formal Methods and Foundations of AI
Chair: Sándor Jenei, C-121

14:00 - 14:20 Levente Bajczi, Dániel Szekeres, Csanád Telbisz and Vince Molnár – *Verifying Concurrent C Programs with Abstraction in Theta*

14:20 - 14:40 Mohammed Khudhair and Attila Fazekas – *A Proposed Method to Reduce the Noise Sensitivity of Binary Classification Based on Robust Deep Neural Networks*

14:40 - 15:00 Richárd Szabó and Dóra Cziborová – *Automata-based Representation of Coordination for Distributed Reactive Systems*

15:00 - 15:20 Milán Mondok, Dániel Szekeres, Levente Bajczi, Dániel Kovács, Mihály Dobos-Kovács and Vince Molnár – *Theta: Modular Model Checker Built For Diverse Algorithm Configurations*

15:20 - 15:40 Ali Ali, Gergely Kovásznai and Mustafa Khaild – *Automated Fair Team Formation in STEAM Activities Using Satisfiability Modulo Theories (SMT)*

15:40 - 16:20 Coffee-break, C-119

Section B
Theoretical Computer Science
Chair: Benedek Nagy, C-121

16:20 - 16:40 Krisztián Mihály, Mónika Kulcsárné Forrai and Gyula Kulcsár – *Solving Multi-Project Scheduling Problems: Efficiency Tests on Transformed Job Shop Problems using a Hybrid Solver*

16:40 - 17:00 Sándor Jenei – *Amalgamation in Classes of Involutive FL_e-algebras*

17:00 - 17:20 Farid Eyvazov and Alwaháb Zoltán – *OpenStack and Kubernetes Integration for Multi-OS Serverless and Microservices*

17:20 - 17:40 Sándor Vályi – *A note on the axiomatizability of some first-order spatio-temporal logics*

17:40 - 18:00 István Pintye, József Kovács and Róbert Lovas – *Fundamental Limitations of Online Supervised Learning in Dynamic Control Loops*

Friday, June 6

08:00 - 13:00 Registration in Main Hall of Building C
Eger, Leányka str. 4., Building C

09:00 - 09:50 **Plenary talk** (Chair: Csaba Biró), C-124
Benedek Nagy – *Nontraditional finite state machines*

09:50 - 10:40 **Plenary talk** (Chair: Csaba Biró), C-124
Bence Tipary – *Tolerance Aspects of Digital Twin-based Autonomous Robotic Solutions*

10:40 - 11:00 Coffee-break, C-119

11:00 - 11:50 **Plenary talk** (Chair: Csaba Biró), C-124
Alin Brindusescu – *Non-invasive Brain Computer Interfaces*

11:50 - 12:40 **Roundtable on AI** (Moderator: Miklós Hoffmann), C-124
Participiants: Benedek Nagy, Tibor Tajti, Péter Molnár

12:40 - 13:00 Poster section (Chair: Károly Nehéz), C-124

13:00 - 14:00 Lunch in Leányka Bisztró in Building C*
*Eger, Leányka str. 4., Building C**

14:00 - 15:40 Sections

15:40 - 16:20 Coffee-break, C-119

16:20 - 18:00 Sections

19:00 - 19:30 We walk from building C to Piknik Étterem (Piknik Restaurant) in Szépasszonyvölgy (Valley of the Beautiful Woman). The distance is about 3 km and 40 minutes walking. (See QR codes!)

19:30 - 24:00 Dinner and wine-sampling in Piknik Étterem

Poster section

Chair: Károly Nehéz, in front of C-124

12:40 - 12:45 Gábor Ujhelyi – *Implementation of Multimodal Course Contents Supported by Generative Artificial Intelligence*

12:45 - 12:50 Lajos Toldi – *Generative AI-Enhanced Adaptive Learning: Integrating GPT Models for Personalized Content and Feedback*

12:50 - 12:55 Sylvert Prian Tahalea and Miklós Krész – *Modularity Aware Label Propagation Community Detection*

Section A
Machine Learning and Data Science
Chair: Károly Nehéz, C-106

14:00 - 14:20 László Kovács and Fadi Atallah – *Explainable AI with Input Feature Analysis*

14:20 - 14:40 Attila Aradi, Péter Takács and Attila K. Varga – *Applications of Machine Learning in Underwater Bioacoustics*

14:40 - 15:00 Mohammed Almurumudhe and Olivér Hornyák – *Motion Enhanced Video Anomaly Detection Using Masked Autoencoder and Hybrid Loss Functions*

15:00 - 15:20 Máté Molnár Norbert and Tibor Tajti – *Local backtracking mechanisms in generative modeling*

15:20 - 15:40 József Gergő Pál and Csaba Biró – *Evaluating Profitability in Sports Betting Using Probabilistic Models and Betting Strategies*

15:40 - 16:00 Gábor Guta and Gábor Kusper – *LLM-Based Framework to Support the Construction of Valid Formal Models*

15:40 - 16:20 Coffee-break, C-119

Section A
Robotics and Digital Twin Technolog
Chair: Bence Tipary, C-106

16:20 - 16:40 Milán Zsolt Bagladi – *Artificial Intelligence for Interpreting Static Human Arm Signals*

16:40 - 17:00 Emin Bayramov and Istenes Zoltán – *Motion-Aware Perception via IMU-Guided Spatial Attention for Autonomous Vehicles*

17:00 - 17:20 András Izsó and István Harmati – *Solving the Team Coordination on Graphs With Risky Edges problem for nonzero self-loop weights*

17:20 - 17:40 Sára Szénási and István Harmati – *Perimeter Defense Game with Nonzero Capture Radius in a Circular Target*"

17:40 - 18:00 Nour Elhouda Ben Saadi and Zoltán Istenes – *Benchmarking Data Logging Strategies in ROS-Integrated Multi-Sensor Robots Under Network Constraints*"

Section B
Intelligent Systems and Smart Manufacturing
Chair: Zoltán Istenes, C-121

14:00 - 14:20 Károly Nehéz and Zoltán Kobolák – *LLMs for Industrial State Detection via Sensor Data and Specifications*

14:20 - 14:40 Anita Agárdi – *Efficiency analysis comparision of the Particle Swarm Optimization and Tabu Search in Flow Shop Scheduling Problem*

14:40 - 15:00 Tarlan Ahadli, Hajder Levente and Kovács Bandó – *Audio Localization for Emergency Vehicles (ALEV): A Deep Denoising Approach for Direction of Arrival Estimation*

15:00 - 15:20 Levente Fazekas and Károly Nehéz – *Multi-objective Genetic and Memetic Algorithms in Flexible Flowshop Scheduling*

15:20 - 15:40 Miklós Krész and Sylvert Prian Tahalea – *Discrete Modeling Approaches in Biomaterials Research*

15:40 - 16:00 Nikola Kovačević and Balázs Dávid – *Reinforcement Learning for Inventory Management in Reverse Supply Chains of Wood*

15:40 - 16:20 Coffee-break, C-119

Section B
Applications of AI in Education
Chair: Sándor Király, C-121

16:20 - 16:40 Gábor Kusper – *We Are Not Afraid of the Wolf! - AI Usage Attitudes Among Hungarian Informatics Students*

16:40 - 17:00 Mihály Dobos-Kovács, András Vörös and Zoltán Micskei – *Beyond Hello World: Teaching Software Engineering with Realistic and Automated Assignments*

17:00 - 17:20 Anikó Apró and Tibor Tajti – *An Adaptive Testing System for Programming Proficiency Using Item Response Theory*

17:20 - 17:40 Péter Tibor Zaletnyik – *Possibilities and Challenges of Automated Question Assessment in Electronic Learning Environments*

Section C
Natural Language Processing
Chair: Tibor Tajti, C-104

14:00 - 14:20 Zijian Győző Yang, Ágnes Bánfi, Réka Dodé, Gergő Ferenczi, Flóra Földesi, Péter Hatvani, Enikő Héja, Mariann Lengyel, Gábor Madarász, Mátyás Osváth, Bence Sárossy, Kristóf Varga, Tamás Váradi, Gábor Prószéky and Noémi Ligeti-Nagy – *PULI Chat: Our First Hungarian Conversational Model*

14:20 - 14:40 Kristof Varga, Péter Hatvani and Zijian Győző Yang – *Full-Parameter Tuning vs. LoRA Tuning on PULI Models*

14:40 - 15:00 Franciska N. Gőz and Erika Baksáné Varga – *Hungarian Case Study on Automated Detection of Body-Shaming Comments Using Machine Learning*

15:00 - 15:20 Erika Baksáné Varga and Attila Baksa – *Syntactic Comparison of Human and AI-Written Scientific Texts*

15:20 - 15:40 Áron Kiss, Károly Nehéz and Olivér Hornyák – *AI-driven Fault Diagnosis from Textual System Logs*

15:40 - 16:20 Coffee-break, C-119

Section C
Isar Tutorial
Chair: Gergely Kovsznai, C-104

16:20 - 18:00 Gergely Buday – *Tutorial of the Isar Structured Proof Language*

Saturday, June 7

08:00 - 11:00 Registration in Main Hall of Building C
Eger, Leányka str. 4., Building C

09:00 - 10:40 Sections

10:40 - 11:20 Coffee-break, C-119

11:20 - 13:00 Sections

13:00 - 13:10 Closing ceremony, C-121

13:10 - 14:00 Lunch in front of C-119

Section A
AI Ethics, Trustworthiness, and Explainability
Chair: Miklós Hoffmann, C-106

09:00 - 09:20 Gyula Balázs Csáki-Hatalovics and Péter Molnár – *Comparative analysis of artificial intelligence regulatory concepts*

09:20 - 09:40 Sándor Király and Ede Troll – *Algorithmic Thinking at Risk? A Case Study on LLM Use Among Computer Science Students*

09:40 - 10:00 Péter Hatvani and Zijian Győző Yang – *Automated Detection of Toxic Comments in Hungarian*

10:00 - 10:20 Ádám Kovács, Roland Gunics and Gergely Kovásznai – *Formally Verifiable Voting Robustness in Neural Network Ensembles*

10:40 - 11:20 Coffee-break, C-119

Section A
Machine Learning and Data Science
Chair: Anikó Apró, C-106

11:20 - 11:40 Martin Kernács and Olivér Hornyák – *LLM-Based Querying and Prognostic Evaluation in Industrial Control Systems*

11:40 - 12:00 János Dávid Balogh and Attila Adamkó – *AI-assisted sensor data processing using machine learning and fuzzy logic*

12:00 - 12:20 Hanna-Georgina Lieb, Tamás Kaszta and Lehel Csató – *Explainable image segmentation with wavelet-network*

12:20 - 12:40 Attila Dobai-Pataky, Balázs Vass and Lehel Csató – *Traffic Prediction in the Energy Sciences Network: A Case Study*

12:40 - 13:00 Gábor Kusper, Erik Zoltán Hidi, Krisztián Kusper, Zijian Győző Yang and Szabolcs Márien – *Applying Tree-Based Convolutional Neural Networks to Classify Design Patterns*

Section B
The Human Brain
Chair: Alin Brindusescu, C-121

09:00 - 09:20 Aurel-Ionuț Coruian, Mihai-David Vuescu, Iosif-Beniamin Goga and Ioana Badiu – *Brain-Computer Interfaces and Virtual Reality- a way of immersion*

09:20 - 09:40 Denis Alexandru Gabriel Andrei, Aurel-Ionut Coruian and Ada Daniela Teodorescu – *Integration of Multiple Brain-Computer Interface Headsets into Collaborative Applications*

09:40 - 10:00 Réka Erdész and Ede Troll – *Making the Boss smarter: A journey from rule-based to learned behaviors*

10:00 - 10:20 Andras Kapy, Alin Brindusescu and Cosmin Bonchis – *Cyber Security Considerations for Active Exoskeletons*

10:40 - 11:20 Coffee-break, C-119

Section B
Intelligent Systems and Smart Manufacturing
Chair: Csaba Biró, C-121

11:20 - 11:40 Flaviu Ioan Gheorghita, Vlad Ioan Bocanet and Laszlo Barna Iantovics – *Fine-Tuning BiomedBERT with LoRA and Pseudo-Labeling for Robust DDI Classification*

11:40 - 12:00 Zsuzsa Simó and Barna Iantovics – *Optimizing Diabetes Prediction: The Role of Feature Engineering and Class Balancing in Machine Learning Models*

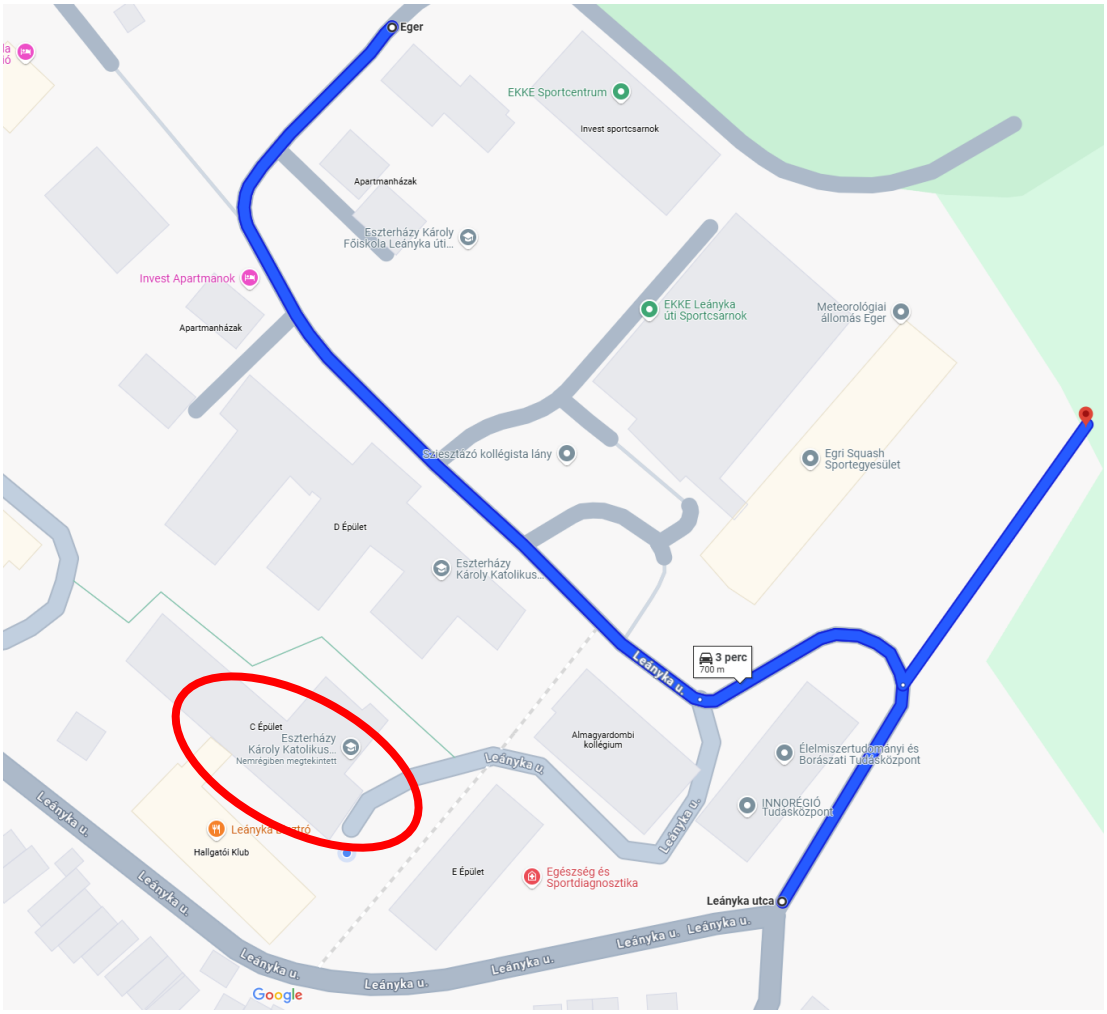
12:00 - 12:20 Daniela Maria Cristea, Ioan Sima, Barna Iantovics and Elisabeta Mihaela Ciortea – *Analysing and Interpreting Information in IoT Using State-of-the-Art RAMI Technologies*

Section C
Practice of the Isar Tutorial
Chair: Gábor Kusper, C-104

09:00 - 10:40 Gergely Buday – *Tutorial of the Isar Structured Proof Language - Practice*

10:40 - 11:20 Coffee-break, C-119

Map



Conference venue (google maps)

Eszterházy Károly Catholic University
Eger, Leányka str. 4., Building C (red ellipse)

Free parking is available along the blue-marked route.

QR codes



Conference venue (google maps)
Eszterházy Károly Catholic University
Building C



Piknik restaurant (facebook)



Piknik restaurant (google maps)
Eger, Szépasszonyvölgy, Éttermek, 5. étterem
Coordinates (latitude, latitude): 47.8911426, 20.3562637

Notes